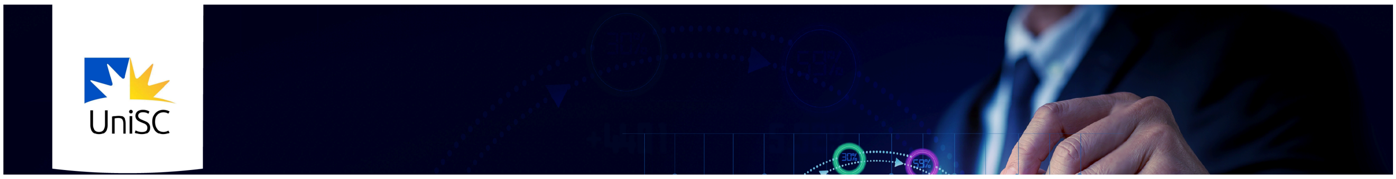
Module 1 Introduction



Introduction to Financial Analytics

What is Financial Analytics?

Financial analytics is the use of data, statistical methods, and analytical tools to analyse, visualise, interpret, and communicate financial information in order to evaluate performance, assess risk, and support financial decision-making.

In practice, financial analytics is applied across a wide range of financial problems, including:

- **Forecasting cash flows**, such as predicting demand, revenues, and expenses to inform budgeting, pricing, and capacity decisions
- **Optimising asset portfolios**, where returns and risks are jointly evaluated to guide investment allocation
- **Evaluating investment performance**, using measures such as return on invested capital (ROIC) and return on investment (ROI)
- **Appraising long-term projects**, using discounted cash flow techniques such as internal rate of return (IRR) and net present value (NPV)
- **Modelling and managing financial risk**, including market risk, volatility, and exposure to macroeconomic and financial shocks

These applications span corporate finance, investment analysis, and risk management, and they rely on quantitative analysis to link financial data to economic and strategic decisions.

What will you be learning this week?

Key Concepts	Content
1.1 Time Series Data in Finance	The role of time in finance and why is time series techniques are important
1.2 Get Started with R Programming Language	What R programming is and how we can get started with it
1.3 R Essentials	The fundamental concepts of R programming language
1.4 Retrieve Stock Prices and Estimate Returns	Writing scripts to access data on stock prices web sources
1.5 Calculate CAGR and Use APIs	Accessing financial and economic time series using APIs and computing compound growth rates

How Module 2 supports your Course Learning Outcomes

This module builds the foundations to work with financial time series data and use R as a practical analytics tool, supporting:

CLO 1: Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.

CLO 2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

CLO 3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

CLO 4: Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

How much time can you expect to spend on Module 1?

This workload is designed to be manageable and flexible (i.e. you don’t need to complete everything in one sitting!).

Estimated learning hours

You will need to spend around 25 hours in total engaging with the learning materials, attending live sessions, studying, and working towards your assessments.